Key Stage 3 Number and Place Value Assessment - **Answers**

1. Write in words:

4453: **Four thousand, four hundred and fifty-three** **[1 mark]**

340 096: **Three hundred and forty thousand and ninety-six** **[1 mark]**

1 290 432: **One million, two hundred and ninety thousand, four hundred and thirty-two** **[1 mark]**

9 034 023: **Nine million, thirty-four thousand and twenty-three** **[1 mark]**

2. Write in figures:

Three thousand, four hundred and eight: **3408** **[1 mark]**

Two hundred and ninety-four thousand, one hundred and twenty-eight: **294 128** **[1 mark]**

Four million, two hundred and twenty thousand: **4 220 000** **[1 mark]**

A quarter of a million: **250 000** **[1 mark]**

3. What is the place value of the 2 in each of the following numbers?

198 299: **hundred** **[1 mark]**

2 040 949: **million** **[1 mark]**

94.542: **thousandth** **[1 mark]**

4. Round to the nearest;

hundred: 5848 = 5800 **[1 mark]**

whole: 1.83 = 2 **[1 mark]**

thousand: 459 900 = 460 000 **[1 mark]**

2 decimal places: 0.07234 = 0.07 **[1 mark]**

5. Use rounding to **estimate** the answers to:

456 × 238 = (500 × 200) = 100 000

(1 mark for sensibly rounded numbers, 1 mark for correct answer with numbers used) **[2 marks]**

584 ÷ 185 = (600 ÷ 200) = 3

(1 mark for sensibly rounded numbers, 1 mark for correct answer with numbers used) **[2 marks]**

6. Insert the correct sign, < or >, in between each pair of numbers:

0 **>** -5 **[1 mark]**

-2 **<** 1 **[1 mark]**

-3 **>** -10 **[1 mark]**

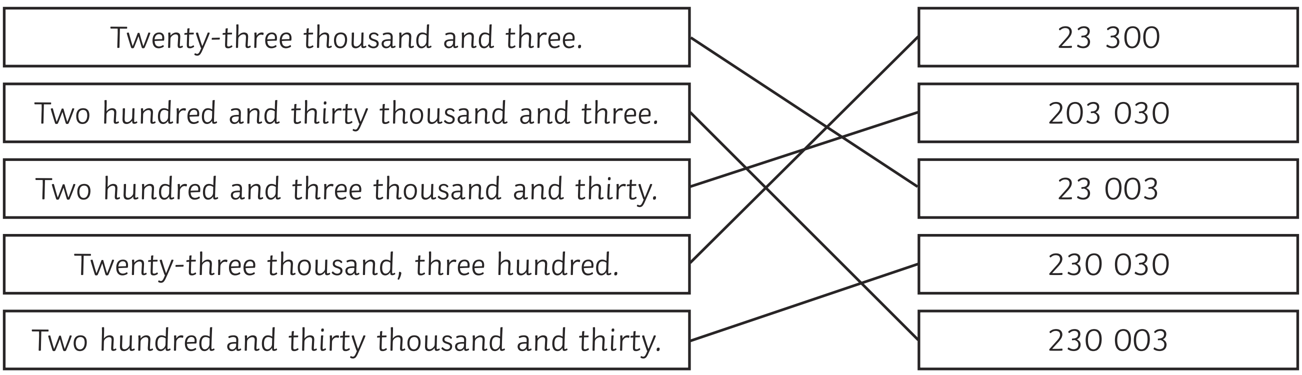
4 × 2 **>** 10 ÷ 2 **[1 mark]**

9 + 5 **<** 20 − 2 **[1 mark]**

0 × 5 **<** 5 + 0 **[1 mark]**

32 + 15 **>** 50 − 4 **[1 mark]**

7. Match each number on the left with one of those on the right.



[3 marks]

(1 mark for 2 correct pairs, 2 marks for 4 correct pairs or 3 marks for all correct)

8. The table shows the local temperature at 9am every day last week. Write the days in order by their temperatures, from coldest to warmest.

Friday, Tuesday, Sunday, Thursday, Saturday, Monday, Wednesday,

(1 mark for 1 or 2 out of place or 2 marks for all correct.)

[2 marks]

9. Use rounding to **estimate** the following:

How many days are there in 17 years?

400×20=8000 **[2 marks]**

(1 mark for sensibly rounded numbers, 1 mark for correct answer with numbers used)

I buy 18 packets of biscuits. They weigh 3070g. What is the mass of one packet of biscuits?

3000÷20=150g **[2 marks]**

(1 mark for sensibly rounded numbers, 1 mark for correct answer with numbers used)

10. The five trees in the park are 2.23m, 3.45m, 1.61m, 2.73m and 1.98m tall. Round each height to 1 decimal place to estimate the total height of the trees in the park.

2.2 + 3.5 + 1.6 + 2.7 + 2.0 = 12m

(1 mark for 4 correctly rounded numbers or 2 marks for correct answer).

[2 marks]

11. a) 2, 3, 7, 11 (all prime numbers identified with no other numbers listed.) **[1 marks]**
   b) 1, 3, 6 (all triangular numbers identified with no other numbers listed.) **[1 marks]**
   c) 64, 125 (both numbers identified with no other numbers listed.) **[1 marks]**
   d) 1, 64 (both numbers identified with no other numbers listed.) **[1 marks]**

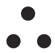
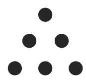
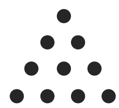
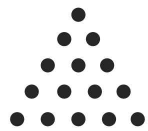
12. 7, 14, 21, 28, 35
(all multiples identified) **[1 marks]**

13. 8 **[1 marks]**

14. $2^2 \times 3^2$
(2 marks awarded for complete answer. Award 1 mark for 2 or 3 identified as factors) **[2 marks]**

15. $2^4 \times 3$ (1 mark for each gap filled correctly) **[2 marks]**

16. a) (1 mark awarded for each complete triangle) **[2 marks]**

1.	2.	3.	4.	5.
				
1	3	6	10	15

b) 55 **[1 marks]**